

MC-EQ and CoverSketch: Destination Privacy via Markov-Chain Equalization in Anonymous DHT Lookups

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Abstract

Destination privacy is the promise that an observer cannot reliably infer *which key* a user is querying. In anonymous overlays, this typically reduces to ensuring that routing transcripts are indistinguishable across destinations. This paper introduces a compact equalization interface for destination privacy: *MC-EQ* (Markov-Chain Equalization) and its deployable “cover set” implementation pattern, *CoverSketch*.

MC-EQ treats routing as a (possibly history-dependent) Markov process over public states (committees, buckets, or routing regions). A mechanism is destination-private if, at each step, its conditional next-state distribution stays within a declared total-variation budget of a public *reference kernel*. CoverSketch realizes this by restricting each state’s outgoing support to a public cover set and equalizing probabilities over that set.

We keep the paper brief by importing the full AnonDHT conditioning model and MUCC plumbing from the state series [1]. Our contribution is the equalization interface, simple composable leakage bounds (TV, ratio/ D_∞ , and session scaling $T \log_2(1 + 2k\delta_{eq})$ under uniform covers), and a receipt-friendly parameterization suitable for the release workflow [2].

Units. Endpoint budgets in this archive are published in *bits* (see the notation/model sheet, Synthesis 8).[19] Accordingly, any conversion from a per-step D_∞ cap to an odds-inflation or MaxL budget is rendered with $\log_2$.

Scope. We define a compact destination-privacy interface for Anonymous-DHT routing transcripts: Markov-Chain Equalization (MC-EQ) and its cover-set implementation pattern (CoverSketch).

Imports. We import the AnonDHT transcript/state conditioning model (including MUCC) from the state series.[1] We treat endpoint conversions (TV $\leftrightarrow$ odds/max-leakage sizing) and release receipt semantics as owned by the companion papers.[4, 2]

Exports. The MC-EQ contract (reference kernel + per-step TV budgets), a deployable CoverSketch construction, and simple composable leakage bounds exportable as receipt parameters.

Artifact policy. Source-only; supporting simulations and kernel-estimation scripts are available on request.[3]

1 MC-EQ: the equalization contract

Let S be a finite set of public routing states (e.g., committees). A lookup for destination d produces a (random) state sequence $X_0, \dots, X_T$. Write the (possibly history-dependent) transition rule as

$$P_d(x_{t+1} \mid h_t) \quad \text{where } h_t = (x_0, \dots, x_t, \text{public side-info}).$$

Definition 1 (Reference kernel). A reference kernel $K(\cdot \mid x)$ is a public Markov kernel on S . It represents the “idealized” destination-independent next-state distribution.

Definition 2 (MC-EQ(ε_t)). A mechanism satisfies **MC-EQ** with parameters $(\varepsilon_t)_{t=0}^{T-1}$ and reference kernel K if for every destination d , every time t , and every history h_t with current state x_t ,

$$\text{TV}(P_d(\cdot \mid h_t), K(\cdot \mid x_t)) \leq \varepsilon_t.$$

The contract is local-in-time and history-robust. It is designed for certification: K and (ε_t) are simple objects that can be published and verified.

1.1 Why TV?

TV is operational: it upper-bounds an observer’s single-shot advantage in distinguishing two distributions. It composes cleanly across steps via standard couplings. For other leakage measures, we can convert TV bounds under mild “floor” assumptions (§2.2).

2 Composed transcript bounds

The main question: what does local MC-EQ buy you for the whole transcript?

2.1 TV composition (coupling view)

Theorem 1 (TV bound for path distributions). Suppose a mechanism satisfies MC-EQ(ε_t) with reference kernel K . Let $\mathcal{L}_d$ be the law of the length- T transcript $(X_0, \dots, X_T)$ for destination d . Then for any two destinations d, d' ,

$$\text{TV}(\mathcal{L}_d, \mathcal{L}_{d'}) \leq \sum_{t=0}^{T-1} \varepsilon_t.$$

Proof sketch. Couple the two processes stepwise. At each step, the conditional next-state laws for d and d' are each within ε_t of the same reference kernel $K(\cdot \mid x_t)$. A maximal coupling shows that the probability the coupled chains diverge at step t is at most ε_t . Union bound over steps yields the claim. $\square$

This TV bound is intentionally simple: it is conservative but receipt-friendly. Sharper bounds (e.g., using spectral contraction when K mixes) can be imported from `[[2026.01.23 - Mathematics: Spectral Anonymity and Optimal Distinguishing Bounds for Random-Walk Delegation in (Possibly Directed) Overlays]]`.

2.2 D_∞ / ratio conversion and session scaling

TV bounds alone do not control likelihood ratios if the reference kernel assigns tiny mass. We therefore make a standard “mass floor” assumption.

Definition 3 (Floor / mass minimum). *For a kernel K , define*

$$q_{\min} = \min\{K(y \mid x) : x \in S, y \in \text{supp}(K(\cdot \mid x))\}.$$

If $K(\cdot \mid x)$ is uniform over a cover set of size $k(x)$, then $q_{\min} = \min_x 1/k(x)$.

Lemma 1 (TV to pointwise ratio under a floor). *Let Q be a distribution with support S and minimum mass $q_{\min} := \min_{y \in S} Q(y) > 0$. If $\text{TV}(P, Q) \leq \delta$, then for all $y \in S$,*

$$Q(y) - 2\delta \leq P(y) \leq Q(y) + 2\delta.$$

Consequently, for all $y \in S$,

$$\frac{P(y)}{Q(y)} \leq 1 + \frac{2\delta}{q_{\min}},$$

and if $\delta < q_{\min}/2$ then also

$$\frac{P(y)}{Q(y)} \geq 1 - \frac{2\delta}{q_{\min}}.$$

Proof. Since $\sum_z |P(z) - Q(z)| = 2\text{TV}(P, Q) \leq 2\delta$, we have $|P(y) - Q(y)| \leq 2\delta$ for every y , yielding the additive bounds. Dividing by $Q(y) \geq q_{\min}$ gives the ratio bounds. $\square$

Per-step ratio and D_∞ . Applying Lemma 1 with $Q(\cdot) = K(\cdot \mid x)$ yields a per-step likelihood-ratio cap:

$$D_\infty(P_d(\cdot \mid h_t) \parallel K(\cdot \mid x_t)) \leq \log_2\left(1 + \frac{2\varepsilon_t}{q_{\min}}\right).$$

In discrete settings, maximal leakage admits simple upper bounds in terms of worst-case likelihood ratios and related max-divergences (see, e.g., [7]). In this series, endpoint conversions and odds-factor interpretations are centralized in the endpoint-bridge note.[4]

Session scaling (uniform support case). A reviewer- and operator-friendly knob is how leakage scales with the number of observable steps T . If the reference kernel is uniform over a cover set of size at most k for every state, then $q_{\min} = 1/k$ and a single-step MC-EQ slack δ_{eq} yields the familiar factor $1 + 2k\delta_{eq}$. Instantiating the series’ generic pointwise-to-session export for per-history likelihood-ratio caps (owned by the scheduling/compiler backbone) gives the additive session bound:

$$\mathcal{L}_{\max}(C \rightarrow O_{1:T}) \leq T \log_2(1 + 2k \delta_{eq}).$$

(For the generic “per-step ratio $\Rightarrow$ session odds” lemma used throughout the archive, see the scheduling/compiler note.[10])

Remark 1 (Sharper bounds via mixing). *The bound above is deliberately conservative and receipt-friendly. If the reference chain mixes rapidly, contraction arguments can replace the linear-in- T envelope with tighter bounds. Those refinements are treated in the spectral anonymity note (wiki) and can be imported here as needed.*

Remark 2 (KL via Pinsker-type bounds). *If desired, TV bounds can be converted to KL using Pinsker-style inequalities (see, e.g., [8]). We avoid committing to a single conversion constant here; the exported interface is TV plus an explicit floor when ratio-style reasoning is required.*

k	δ_{eq}	bound at $T = 50$ (bits)
8	0.005	5.55
8	0.010	10.71
8	0.020	20.03
16	0.005	10.71
16	0.010	20.03
16	0.020	35.68
32	0.005	20.03
32	0.010	35.68
32	0.020	59.45

Table 1: Analytic maximal-leakage upper bound $T \log_2(1 + 2k\delta_{eq})$ for representative parameters (uniform k -support reference).

3 CoverSketch: a deployable pattern

MC-EQ is a contract; CoverSketch is a pattern for realizing it. The key insight is to choose a public cover support at each state and then equalize within that support.

3.1 Public cover sets

For each state $x \in S$, choose a public cover set $\mathcal{C}(x) \subseteq S$. The reference kernel is typically

$$K(y \mid x) = \begin{cases} 1/|\mathcal{C}(x)| & y \in \mathcal{C}(x), \\ 0 & \text{otherwise.} \end{cases}$$

This makes the floor parameter $\alpha = \min_x 1/|\mathcal{C}(x)|$ explicit.

3.2 Equalization with slack

A mechanism implements MC-EQ by ensuring its actual conditional next-state law stays within ε_t of this uniform reference. Operationally, that slack is used to (i) respect feasibility constraints (committee availability), and (ii) adapt to transient performance goals.

3.3 A minimal CoverSketch certificate

For certification and release receipts [2], CoverSketch exports:

- the public state set S and cover sets $\mathcal{C}(x)$;
- the induced kernel K and floor α ;
- the per-step (or per-phase) slack schedule (ε_t) ;
- any conditioning rules used to define S (imported from [1]).

These are small, stable objects compared to raw traces.

4 Design tradeoffs

Sparsity vs. leakage. Smaller cover sets reduce cost but decrease α and therefore worsen D_∞ bounds. This makes the tradeoff explicit and auditable.

Approach	What it targets	Session risk / mismatch
Anonymous-path DHT lookups (e.g., <i>Octopus</i>) [15]	Initiator anonymity and robustness against active attacks	Destination leakage can persist via history-dependent scheduling and retries unless equalization is explicit
OT/PIR-style query privacy [16, 17]	Hides the queried key from intermediates/servers (crypto assumptions)	Does not by itself equalize which committees/regions are contacted or control-plane patterns
Query obfuscation (double hashing / prefix fetching) [18]	Reduces interest leakage to individual responders	Repetition may still amplify bias; conditioning on paths and retries can accumulate evidence
MUCC-style marginal equalization [1]	Makes single-step contact distribution less distinguishing	Not stable under history-conditioned schedulers without an explicit contract
MC-EQ (this paper)	History-conditioned equalization to a public Markov reference	Designed to remain meaningful under sessions; leakage scales with explicit (T, k, δ_{eq}) (Table 1)

Table 2: Baselines and how they relate to session adversaries.

Mixing helps. If the reference chain K mixes quickly, then even conservative TV union bounds can be improved using contraction arguments [6]. In this series, those sharper bounds are developed in the spectral anonymity note (wiki).

Congestion coupling. Cover sets constrain routing choices and can interact with congestion. We treat congestion observables separately via Congestion-EQ [5]; the release workflow should export both certificates and treat them as linked budgets.

5 Baselines and what they protect (discussion)

Table 2 situates MC-EQ relative to common DHT privacy tactics. The intent is not to survey the space (see the related-work routing map for that), [11] but to make explicit *which witness channel* each baseline addresses.

6 Deployment integration and receipt exports

MC-EQ/CoverSketch are designed to export verifiable line items for the release-receipt workflow.[2] A minimal deployment should export:

- equalization slack δ_{eq} (per step or per phase) and a reference-kernel identifier;
- a support floor $q_{\min}$ (equivalently a cover-size bound $k_{\max}$ when using uniform covers);
- the mixing/cover horizon m and an explicit per-step contact budget k (or committee fanout);
- public lookup parameters that shape transcript length (e.g., concurrency/termination thresholds in iterative Kademlia-like deployments);
- routing-table constraints that define the effective overlay graph (filters, admissible peers/committees); and

- any retry/session-horizon assumptions used to translate per-step bounds into window budgets (Table 1).

Exporting truncation vs. implementation deviation. CoverSketch typically targets a truncated row $\tilde{R}_x$ (a top- k or cover-restricted approximation of an ideal reference row R_x). A conservative receipt should therefore separate (i) a *topology approximation* term $\eta_x := \text{TV}(\tilde{R}_x, R_x)$ and (ii) a *mechanism deviation* term δ_{impl} capturing how far the realized scheduler is from $\tilde{R}_x$ under churn, packet loss, and implementation details. By triangle inequality,

$$\text{TV}(P_d(\cdot \mid h_t), R_{x_t}) \leq \delta_{\text{impl}} + \eta_{x_t}.$$

This keeps the bound honest without forcing the paper to model every deployment detail.

Churn and retries as explicit error terms. In practice, the realized transition kernel deviates from the intended reference due to churn (committee membership changes), packet loss, and retry logic. A tight export is therefore an *effective* per-step slack $\delta_{eq}^{\text{eff}} = \delta_{alg} + \delta_{churn} + \delta_{retry}$, where δ_{churn} can be estimated by comparing observed transition frequencies to the advertised public topology over rolling windows. Treating these as first-class receipt line items keeps the bound auditable.

Bundle record (optional). In the release workflow, MC-EQ/CoverSketch evidence can be summarized by an optional `mceq.json` record that commits to: the public topology snapshot (e.g., a graph hash), the reference-kernel parameters, chosen (m, k) , per-state cover sets $\mathcal{C}(x)$, and tail masses η_x . Downstream verifiers can replay the cover construction and check that the exported $(\delta_{eq}, q_{\min})$ are realizable.

Engineering note. Even when destination equalization is correct, other control-plane surfaces may dominate leakage (indexers, gateways, provider advertisements, and monitoring beacons). The release receipt should therefore treat MC-EQ as one line item among several and compose it with the companion tiers (retries/stop times, congestion witnesses, and control-plane releases).[12, 13, 14]

7 Conclusion

MC-EQ makes destination privacy modular: it is a local TV contract against a public reference chain. CoverSketch is a simple, certificate-friendly pattern that instantiates that contract with explicit sparsity-leakage parameters. Together, they let deployments publish destination privacy as a checkable release artifact, rather than an informal argument.

8 Takeaways

- **Destination privacy needs its own witness:** even if queried keys are protected, *where* traffic exits (destinations, providers, gateways) can be key-correlated without explicit accounting.
- **MC-EQ/CoverSketch is a bridge:** treat destination leakage as an equalization problem over a destination sketch, so knobs (cover, mixing, committee choices) compile to a TV-style bound.
- **Export line items for receipts:** the goal is a small manifest + accountant transcript that can be replayed and anchored in release receipts, not a narrative argument.

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